

EX.No: 6-EDGE DETECTION

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```
In [3]: import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('cat.jpg') # Replace with your image path
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Original Image
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

```
Out[3]: (np.float64(-0.5), np.float64(473.5), np.float64(265.5), np.float64(-0.5))
```

Original Image



```
In [4]: sobel_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=5) # Sobel in x direction
sobel_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=5) # Sobel in y direction
sobel_combined = cv2.magnitude(sobel_x, sobel_y) # Combine both directions
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')
```

```
Out[4]: (np.float64(-0.5), np.float64(473.5), np.float64(265.5), np.float64(-0.5))
```

Sobel Edge Detection



```
In [5]: laplacian = cv2.Laplacian(gray_image, cv2.CV_64F)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian Edge Detection')
plt.axis('off')
```

```
Out[5]: (np.float64(-0.5), np.float64(473.5), np.float64(265.5), np.float64(-0.5))
```

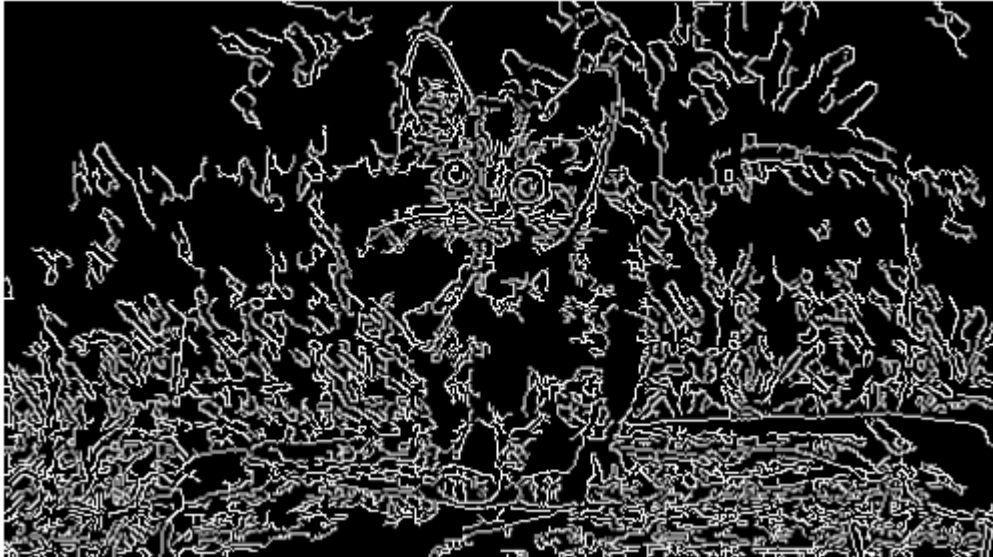
Laplacian Edge Detection



```
In [6]: canny_edges = cv2.Canny(gray_image, 50, 150)
plt.imshow(canny_edges, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off')
```

```
Out[6]: (np.float64(-0.5), np.float64(473.5), np.float64(265.5), np.float64(-0.5))
```

Canny Edge Detection



In []: